

EMS — Phase 1

Wireframes & Screen Specification

Employee Management System · Phase 1 (Pre-Payroll)

Document type: low-fidelity wireframes with annotations

Contents

01 Login	06 Dashboard — Employee	11 Attendance — Records
02 Forgot Password	07 Employees — List	12 Leave — Requests & Approvals
03 OTP Verification	08 Employees — Profile	13 Holiday Calendar
04 Dashboard — HR Admin	09 Employees — Create / Edit	14 Permissions Matrix
05 Dashboard — Manager	10 Departments	15 Settings

01 Login

[Logo] EMS

Sign in to your workspace

Use your work email to continue

Email

you@company.com

Password

••••••••

[Forgot password?](#)

Sign in

[Sign in with SSO \(phase 2\)](#)

Annotations

Module

- Auth

Render

- Static / RSC. No data required.
- Form is a Client Component (React Hook Form + Zod).

Components used

- AuthShell layout
- DynamicForm (single-section variant) OR direct form
- Primary button, text input, link

Behavior

- Submits to POST /api/v1/auth/login
- On 200: store access token in memory, refresh token in httpOnly cookie
- Redirect to /dashboard or to ?next= param if present
- On 401: inline error 'Invalid credentials' under the form

Validation

- Email format, password min 8 chars (server enforces real rules)

Edge cases

- Rate limit hit (429): show banner with retry-after
- Account disabled: distinct message
- MFA required: redirect to /otp-verification

A11y

- Labels associated to inputs
- Error messages tied to inputs via aria-describedby
- Submit button shows loading state during request

02 Forgot Password

[Logo] EMS

Forgot your password?

We will send a reset link to your email.

Email

you@company.com

Send reset link

[← Back to sign in](#)

Annotations

Render

- Static. Form is a Client Component.

Behavior

- POST /api/v1/auth/forgot-password { email }
- Always returns 202 (do not leak whether email exists)
- Show success screen: 'If an account exists, we have sent a link.'
- Server enqueues an email job via BullMQ

Anti-abuse

- Rate limit: 3 requests / hour / IP
- Captcha after 2nd attempt in a session

Email content

- Tokenized link valid 30 minutes, single-use
- Token stored hashed in DB

Reset flow (next screen, not pictured)

- GET /reset-password?token=... validates token
- User sets new password (min 12 chars, complexity rules)
- All existing sessions revoked on success

03 OTP Verification

[Logo] EMS

Verify it's you

Enter the 6-digit code sent to your email.

Verify and continue

Didn't get the code? Resend in 0:45

Annotations

When this screen appears

- Only if MFA is enforced for the user (Phase 2-ready hook)
- On suspicious login (new device / new country)

Component

- Custom OTP input (6 cells, auto-advance, paste-aware)

Behavior

- POST /api/v1/auth/verify-otp { code, challengeld }
- Issues access + refresh tokens on success
- Lockout after 5 failed attempts; user must restart login

Resend

- Throttled: 60s cooldown, max 3 resends per challenge

Edge cases

- Code expired (10 min TTL): clear inputs, show inline message
- Tab away and return: timer continues, inputs preserved if same challenge

04 Dashboard — HR Admin

[Logo] EMS

Dashboard

Search...

Aman K.

Dashboard

Employees

Departments

Attendance

Leave

Holidays

Permissions

Settings

Home

Welcome back, Aman

Add Employee

Total Employees

1,240

↑ 12 vs last month

Active Today

1,087

↑ 3.1%

On Leave

84

↓ 2

Open Requests

23

5 urgent

Attendance — last 30 days

Headcount by Department

Engineering

Sales

Ops

HR

Recent Activity

Who	Action	Resource	When
Priya S.	created	Employee #E2041	20m ago
Priya S.	approved leave	LR-1187	10m ago
Priya S.	updated	Department: Sales	5m ago
Priya S.	uploaded doc	Aadhaar.pdf	24m ago
Priya S.	deleted	Role: Intern	31m ago

User Menu

Annotations

- Render
- RSC for initial paint; stats fetched on the server
 - Streaming Suspense boundary per card so slow queries don't block fast ones
 - Charts and table hydrate as Client Components

- Components
- StatsCard × 4
 - ChartEngine (line/bar)
 - ChartEngine (donut)
 - ActivityTimeline / DynamicTable in compact mode

- Data sources
- GET /api/v1/analytics/summary
 - GET /api/v1/analytics/attendance?range=30d
 - GET /api/v1/analytics/headcount-by-department
 - GET /api/v1/audit-logs?limit=10

- Permissions
- analytics:read on summary endpoints
 - Super Admin sees a tenant switcher in the topbar

- Performance
- Summary endpoint cached 60s in Redis
 - Each card has its own Suspense fallback (skeleton)

- Empty / error
- Each card independently shows empty / error state
 - Page does not fail wholesale if one widget errors

05 Dashboard — Manager

[Logo] EMS

Dashboard

Employees

Departments

Attendance

Leave

Holidays

Permissions

Settings

User Menu

Search...

Aman K.

Home

My Team

Bulk approve

View team

Team size

14

Present today

11

↑ 1

Pending approvals

5

3 leave, 2 reg.

Avg. attendance

94%

this month

Priya S.

Annual leave · 3 days

Deny

Approve

Rahul M.

Sick leave · 1 day

Deny

Approve

Anita D.

WFH · 5 days

Deny

Approve

Vikram P.

Comp off · 1 day

Deny

Approve

Team Attendance — This Week

	M	T	W	T	F
P. Sharma	P	P	L	P	A
R. Mehta	P	L	P	A	P
A. Das	L	P	A	P	P
V. Patil	P	A	P	P	L
S. Iyer	A	P	P	L	P

Annotations

- Role scope**
- Manager sees only their direct reports (recursive option in Phase 2)
 - Permissions: team:read, leave:approve (their team only)

- Approval actions**
- Approve / Deny mutate via PATCH /api/v1/leave-requests/id
 - Optimistic update with rollback on error
 - Bulk approve opens a ModalEngine with multi-select

- Attendance grid**
- Server returns aggregated cells; client just renders
 - P = Present, A = Absent, L = Leave, W = WFH, H = Half day

- Notifications**
- Real-time updates via SSE on /events stream
 - New request → toast + counter increment without refresh

- Empty states**
- 'No pending approvals' with illustration when queue is empty
 - Empty team grid with a CTA to define team in Settings

06 Dashboard — Employee

[Logo] EMS

Dashboard

Dashboard

Employees

Departments

Attendance

Leave

Holidays

Permissions

Settings

User Menu

Search...

Aman K.

Home

Hi Priya

Request leave

Today's attendance

09:14 AM

Checked in · WFH

Check out

Leave balance

6

Casual

4

Sick

View history

Upcoming holidays

25 Jun

Eid

15 Aug

Independence Day

02 Oct

Gandhi Jayanti

My Documents

Offer Letter.pdf

Active

Aadhaar.pdf

Verified

PAN Card.pdf

Verified

BankAcc.pdf

Pending

My Team

Aman K. — Manager

Engineering

Rahul M.

Engineering

Anita D.

Engineering

Vikram P.

Engineering

Annotations

Scope

- Visible to every authenticated user as their landing screen
- Different content from HR/Manager — RoleSwitch decides which
- dashboard renders

Components

- AttendanceClockCard
- LeaveBalanceCard
- HolidayListCard
- EmployeeDocumentList
- TeamMiniList

Data

- GET /api/v1/attendance/today
- GET /api/v1/leave/balance/me
- GET /api/v1/holidays/upcoming?limit=3
- GET /api/v1/employees/me/documents
- GET /api/v1/employees/me/team

Behavior

- Check-in/out mutate /api/v1/attendance/check-in (with geo if enabled)
- Request leave button opens drawer with the leave form

Reduced motion

- No animation on the clock display when prefers-reduced-motion is set

07 Employees — List

[Logo] EMS

Dashboard

Employees

Departments

Attendance

Leave

Holidays

Permissions

Settings

User Menu

Employees

Search...

Aman K.

Home / Employees

Export

Add Employee

Search by name, code, email...

Department

Save view

1,240 employees

Columns

Density

☐	Priya Sharma priya@...	E2040	Engineering	Sr Engineer	Active
☐	Rahul Mehta rahul@...	E2041	Sales	AM	Active
☐	Anita Das anita@...	E2042	Ops	Lead	On Leave
☐	Vikram Patil vikram@...	E2043	HR	HRBP	Active
☐	Suresh Iyer suresh@...	E2044	Engineering	Engineer	Inactive
☐	Neha Gupta neha@...	E2045	Engineering	Sr Engineer	Active
☐	Arjun Kapoor arjun@...	E2046	Sales	AM	Active
☐	Divya Rao divya@...	E2047	Ops	Lead	On Leave
☐	Kabir Singh kabir@...	E2048	HR	HRBP	Active
☐	Meera Jain meera@...	E2049	Engineering	Engineer	Inactive

Showing 1–10 of 1,240

< 1 2 3 ... 124 >

Annotations

Engine

- DynamicTable<Employee> with columns array
- Columns: checkbox, avatar-text, text, text, text, badge, date, actions
- Virtualization auto-on (>100 rows)

Filters

- FilterEngine bound to URL params via nuqs
- Saved views per user (FilterEngine.savedViews)

Bulk actions

- Selected rows enable: Bulk export, Bulk deactivate (HR only)
- Confirmation via ModalEngine

Row click

- Navigates to /employees/:id (profile)

Permissions

- Add Employee button gated by employees:write
- Manager sees only their team (server filters by reports_to)
- Export gated by employees:export

Performance

- Server pagination (offset). Cursor option for very large tenants
- Column visibility persisted per user in localStorage
- Search debounced 300ms

Empty / error

- EmptyState: 'No employees yet — Add your first one'
- ErrorState with retry on fetch failure

08 Employees — Profile

[Logo] EMS

Dashboard

Employees

Departments

Attendance

Leave

Holidays

Permissions

Settings

User Menu

Employees

Search...

Aman K.

Employees / Priya Sharma

Priya Sharma

Deactivate

Edit

Priya Sharma

Senior Engineer · Engineering · E2041

Active

Full-time

Joined 12 Mar 2022

Reports to Aman Kumar

OverviewJobDocumentsAttendanceLeaveActivity

Personal

Date of birth14 Aug 1992

GenderFemale

Phone+91 98xxxxxx21

Personal emailpriya.s@gmail.com

AddressSector 18, Noida, UP

Emergency contact+91 98xxxxxx70

Job

DesignationSenior Engineer

EmploymentFull-time

ManagerAman Kumar

DepartmentEngineering

LocationDelhi

Pay currencyINR

Documents

Offer Letter.pdf

Aadhaar.pdf

PAN.pdf

View all

Recent activity

Approved leave LR-11872d ago

Uploaded Aadhaar.pdf1w ago

Role changed → Senior3m ago

Joined Engineering2y ago

Annotations

Render

- Header + read-only sections: RSC
- Tabs: Client component, switches without route change (query param ?tab=)
- Each tab can lazy-load via next/dynamic

Components

- Header card with avatar + chips
- Tabs (shadcn)
- Key-value pair grid (reusable as KeyValueGrid)
- Documents preview list (EmployeeDocumentList)
- ActivityTimeline

Permissions

- Self-view: always allowed
- Manager: read team, write limited (designation, location only)
- HR Admin: full read/write except sensitive financial (Phase 2)
- Edit button hidden if user lacks employees:write

Data

- GET /api/v1/employees/:id
- GET /api/v1/employees/:id/documents?limit=3
- GET /api/v1/audit-logs?entity=employee&id=:id&limit=4

Edit flow

- Edit button → /employees/:id/edit (separate page) OR inline
- DrawerEngine
- Phase 1: separate page for simplicity

Deactivate

- Triggers ConfirmDialog ('Type employee code to confirm')
- Soft delete sets deleted_at; revokes sessions

09 Employees — Create / Edit

[Logo] EMS

Dashboard

Employees

Departments

Attendance

Leave

Holidays

Permissions

Settings

User Menu

Employees

Search...

Aman K.

Employees / Add

Cancel

Save

1. Personal

2. Job

3. Documents

4. Access

Identity

First name *

Priya

Last name *

Sharma

Work email *

priya@acme.com

Personal email

priya.s@gmail.com

Phone

+91 98xxxxxxx

Date of birth

14 / 08 / 1992

Job

Employee code *

Auto / E20XX

Desianation *

Senior Engineer

Department *

Engineering

Re...

Ar...

Save draft

Save & continue

Joined on *

12 / 03 / 2022

Emplovmnt tvpe *

Full-time

Annotations

Engine

- DynamicForm driven by employeeCreateSchema (Zod)
- Sections array: Identity, Job, Documents, Access

Schema sharing

- Schema lives in packages/types/src/schemas/employee.schema.ts
- Imported by both frontend and backend — single source of truth

Behavior

- Validates on blur and on submit
- Server errors (422) mapped back to field errors via form.setError
- Auto-save draft to localStorage every 30s (keyed by user + form id)
- Optional 'Generate code' button calls /api/v1/employees/next-code

Async fields

- Department dropdown: lazy fetch via /api/v1/departments?fields=id,name
- Manager dropdown: searchable, server-side filter

Documents step

- FileUploader with virus-scan status
- Pre-signed S3 upload from client, server records metadata

Access step

- Select role from tenant roles
- Toggle: 'Send invite email' (default on)

Phase 2 hooks

- Salary fields slot in here in Job section, behind payroll:write permission

[Logo] EMS

Dashboard

Employees

Departments

Attendance

Leave

Holidays

Permissions

Settings

User Menu

Departments

Search...

Aman K.

Home / Departments

Departments

Add department

Ora tree

Search departments...

■ Acme Corp (root)1,240

■ Engineering412

Frontend98

Backend144

DevOps32

■ Sales210

■ Operations180

■ HR24

Recruiting12

People Ops12

■ Finance44

Engineering

Headcount412

Sub-depts3

Managers12

Employees in this department

Aman K. — ManagerLead

Priya S.Sr. Eng

Rahul M.Eng

Anita D.Eng

Vikram P.Eng

Neha G.Intern

Annotations

- Data model
- Nested via parent_id self-reference
 - Server returns flat list with parent_id; client builds tree (memoized)
 - Depth limit enforced (e.g. 5) to keep UI sane

- Engine
- Custom DepartmentTree component (not a generic engine — tree is dept-specific)
 - Right side uses KeyValueCollection + DynamicTable (compact)

- Behavior
- Click node → loads detail (URL: /departments/:id)
 - Add department: opens DrawerEngine (form: name, parent, head)
 - Drag-to-reparent: Phase 2

- Validation
- Cannot create cycles (server check)
 - Cannot delete dept with active employees (offers 'reassign' flow)

- Performance
- Tree is fetched once and cached aggressively (5 min)
 - Headcounts are aggregated server-side

11 Attendance — Records

[Logo] EMS

Dashboard

Employees

Departments

Attendance

Leave

Holidays

Permissions

Settings

User Menu

Attendance

Search...

Aman K.

Home / Attendance

Attendance

Export

Regularize

June 2025

Department: All

Employee

Present

Priya Sharma — June 2025

Mon	Tue	Wed	Thu	Fri	Sat	Sun
			1 P	2 P	3	4
5 WFH	6 P	7 P	8 L	9 P	10	11
12 P	13 L	14 P	15 L	16 P	17	18
19 P	20 P	21 WFH	22 WFH	23 P	24	25
26 P	27 P	28 L	29 WFH	30 P		

Month summary

Present	18
WFH	3
Leave	2
Absent	1
Holidays	1
Avg. hours / day	8h 42m
Late check-ins	2

Regularization requests

12 Jun · Pending out

08 Jun · Approved k in

03 Jun · Pending

Annotations

- Two views
- Calendar (this) + Table (toggle in toolbar)
 - Switch is a tab control; URL: ?view=calendar|table
- Data
- GET /api/v1/attendance?employeeId=&month=YYYY-MM
 - Returns daily records aggregated server-side
- Scope
- Employee: sees only own records
 - Manager: dropdown to any team member
 - HR Admin: dropdown to any employee in tenant
- Regularize flow
- Opens DrawerEngine with date picker + reason + supporting doc
 - Manager receives notification + appears in their approval queue
- Cell click
- Day cell click opens day-detail Drawer: clock-in/out, hours, location, notes
- Export
- CSV / XLSX export of month for selected scope (BullMQ job, emailed when ready)
- Edge cases
- Future dates greyed out and not interactive
 - Holidays show holiday name in tooltip and a small icon

12 Leave — Requests & Approvals

[Logo] EMS

Dashboard

Employees

Departments

Attendance

Leave

Holidays

Permissions

Settings

User Menu

Leave

Search...

Aman K.

Home / Leave

Leave

Request leave

My Requests (4)

Approvals (5)

Team Calendar

Balances

5 pending approvals

Bulk approve

Bulk deny

☐	Priya S.	Annual	24 Jun – 26 Jun	3	Family wedding
☐	Rahul M.	Sick	20 Jun	1	Fever
☐	Anita D.	WFH	23 Jun – 27 Jun	5	Maintenance at hom
☐	Vikram P.	Comp off	21 Jun	1	Worked weekend
☐	Suresh I.	Annual	01 Jul – 05 Jul	5	Personal trip

Annotations

Tabs

- My Requests: employee's own history
- Approvals: manager queue (their team only)
- Team Calendar: month view of who is on leave when
- Balances: per-leave-type balance for self or selected employee

Engine

- DynamicTable<LeaveRequest> with selection mode multi
- Row action buttons render via type: 'actions' columns + permission gate

Behavior

- Approve / Deny: PATCH /api/v1/leave-requests/:id { status, comment? }
- Bulk action opens ModalEngine to collect optional shared comment
- Optimistic UI; on error, row reverts and toast shows reason

Conflicts

- Server detects overlapping team leave and warns approver
- Warning chip appears next to date when team coverage < threshold

Notifications

- Requester gets in-app + email on decision
- Approval pulls live via SSE; counter in tab updates without refresh

Audit

- Every state transition writes to audit_logs with actor and reason

13 Holiday Calendar

[Logo] EMS

Dashboard

Employees

Departments

Attendance

Leave

Holidays

Permissions

Settings

User Menu

Holidays

Search...

Aman K.

Home / Holidays

Holiday Calendar — 2025

Import .ics

Add holiday

Jan

1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30

Feb

1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30

Mar

1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30

Apr

1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30

May

1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30

Jun

1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30

Jul

1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30

Aug

1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30

Sep

1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30

Oct

1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30

Nov

1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30

Dec

1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30

Annotations

Views

- Year overview (this) is default
- Click a month → expanded month view with names/details
- Toggle list view for plain-text export

Data

- GET /api/v1/holidays?year=2025
- Cached aggressively (1h) — holidays rarely change

Multi-location

- Tenant can define location-scoped holidays
- Employee sees holidays for their assigned location only
- HR Admin sees union with location filter

Add holiday

- DrawerEngine form: name, date, locations[], optional/mandatory
- Import: parse .ics file via BullMQ job, preview before commit

Conflicts

- If a holiday overlaps an approved leave, the leave automatically converts to a half day or full WFH per tenant rule (Phase 1 keeps it manual: just flag it)

14 Permissions Matrix

[Logo] EMS

Dashboard

Employees

Departments

Attendance

Leave

Holidays

Permissions

Settings

User Menu

Permissions

Search...

Aman K.

Home / Permissions

Roles & Permissions

Add role

Save changes

	Super Admin	HR Admin	Manager	Employee	Auditor
Employees					
Read employees:read	✓	✓	✓	✓	✓
Create / edit employees:create	✓	✓			
Delete employees:delete	✓	✓			
Export employees:export	✓	✓			
Departments					
Read departments:read	✓	✓	✓		✓
Create / edit departments:create	✓	✓			
Attendance					
Read attendance:read	✓	✓	✓	✓	✓
Edit / regularize attendance:write	✓	✓	✓		
Leave					
Read leave:read	✓	✓	✓	✓	✓
Request leave:request	✓	✓		✓	
Approve / deny leave:approve	✓	✓	✓		
Permissions					
Manage roles & permissions permissions:manage	✓				
Audit					
Read audit logs audit:read	✓	✓	✓	✓	✓

Annotations

- Scope

 - Super Admin only. Other roles do not see this menu item.
- Model

 - roles (per tenant) × permissions (global catalog) joined by
 - role_permissions
 - user_roles assigns roles to users
 - Built into CASL ability on login or fetched via /auth/me
- Behavior

 - Toggle checkboxes mark the form dirty
 - Save Changes button enabled only when dirty
 - Confirm dialog warns: 'X users will lose / gain access'
 - On save: bulk PATCH /api/v1/roles/:id/permissions
- Safety

 - Super Admin role cannot have permissions:manage unchecked
 - Last Super Admin user cannot be downgraded (server enforced)
- Custom roles

 - HR Admin can create new roles in their tenant (e.g. 'Recruiter')
 - Permission catalog is read-only (defined by the product)
- Audit

 - Every permission change writes to audit_logs with diff and actor

15 Settings

[Logo] EMS

Settings

Dashboard

Employees

Departments

Attendance

Leave

Holidays

Permissions

Settings

User Menu

Search...

■ ● Aman K.

Home / Settings

Settings

WORKSPACE

Company profile

Branding

Locale & timezone

Working hours

PEOPLE

Leave types

Holiday calendar

Attendance rules

SECURITY

Authentication

Sessions & devices

Audit log

NOTIFICATIONS

Email templates

In-app preferences

INTEGRATIONS

Email (SES / Resend)

Storage (S3)

Webhooks

BILLING

Plan

Invoices

Company profile

Public-facing information about your workspace.

Logo

Upload

PNG or SVG, max 1 MB

Legal name

Acme Corporation Pvt Ltd

Display name

Acme

Country

India

Default currency

INR

Timezone

Asia/Kolkata (UTC+05:30)

Fiscal year start

01 April

Primary contact email

hr@acme.com

Support phone

+91 11 4xxxxxxx

Discard

Save

Annotations

- Pattern**
- Two-column: left nav (groups + items), right detail panel
 - Each item is a sub-route: /settings/workspace/company, /settings/security/auth, ...
- Permission gating**
- Each section item is wrapped in PermissionWrapper
 - Items the user cannot access are hidden entirely (not greyed out)
- Audit**
- Every save writes an audit entry with diff (old vs new values)
- Dirty tracking**
- Save / Discard appear only when form is dirty
 - Browser warns on unload if dirty
- Locale switching**
- Saving timezone triggers cache invalidation for date-displaying queries
- Future hooks**
- Payroll settings (pay schedule, tax provider, statutory components) plug into 'People' group in Phase 2
 - SSO settings appear under 'Security' when Phase 2 ships